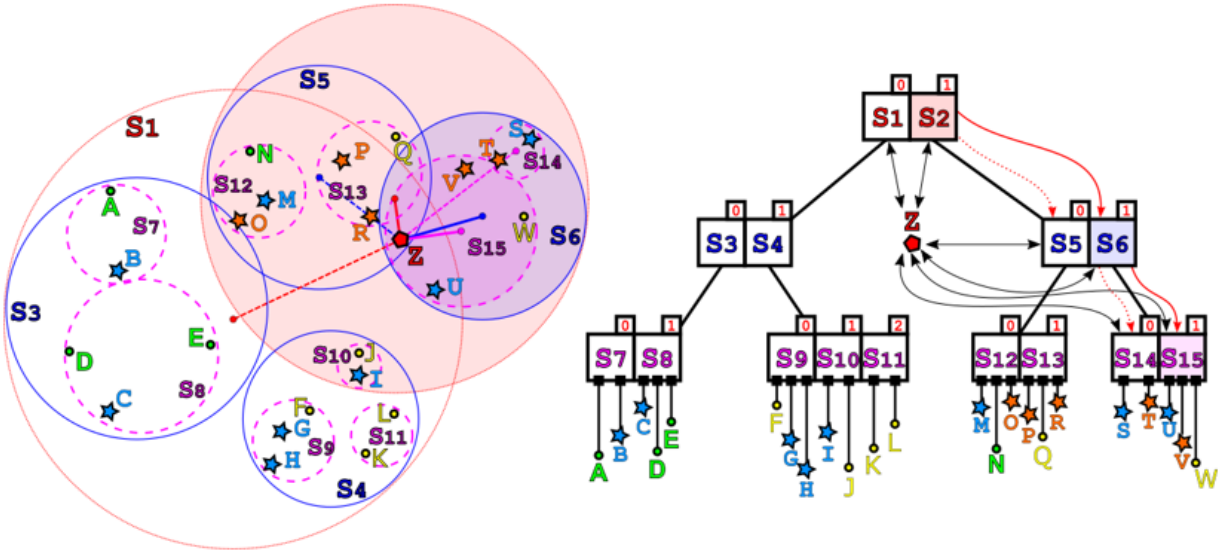


Figure 10.12 An example where the point Z, that will be inserted into the tree, is added to the closest leaf, S_{14}, whose bounding envelope becomes larger as a result, and overlaps another existing leaf, S_{15}, that could have held Z within its bounding envelope. On the right, notice how S_{14}'s centroid moves as a result of adding the new point to the sphere.

On the other hand, though, using this heuristic greatly simplify the code and improve the running time: this way we have a worst-case bound (for this operation) of `O(logm(n))`, because we only follow one path from root to a leaf. If we were to traverse all branches intersecting Z, in the worst case we could be forced to visit all leaves.

Moreover, the code would also become more complicated, because we might have to handle differently the cases where no leaf, exactly one leaf, or more than one leaf intersecting Z are found.